Experiment No. 5

Pulse Shaping and Matched Filtering using Root Raised Cosine Filter

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Aim

To study pulse shaping and matched filtering using a Root Raised Cosine (RRC) filter in a QPSK communication system implemented using GNU Radio and PlutoSDR.

Objective

The objective of this experiment is to reduce inter-symbol interference (ISI) using pulse shaping techniques and to analyze the effect of Root Raised Cosine filtering on the transmitted QPSK signal.

Theory

In digital communication systems, abrupt symbol transitions produce large spectral side lobes and cause inter-symbol interference (ISI). Pulse shaping is used to limit the bandwidth of the transmitted signal and reduce ISI.

A Root Raised Cosine (RRC) filter is widely used for pulse shaping because it satisfies the Nyquist criterion for zero inter-symbol interference.

The overall response of the transmitter and receiver filters forms a Raised Cosine response:
[H(f)=H$_{RRC}(f) \times H_{RRC}(f)$]
where (H$_{RRC}(f)) representstheRootRaisedCosinefilterresponse.$

The roll-off factor α controls the excess bandwidth of the transmitted signal. Lower values provide better bandwidth efficiency, while higher values improve timing recovery.

At the receiver, an identical Root Raised Cosine filter acts as a matched filter, maximizing signal-to-noise ratio and minimizing inter-symbol interference.

Equipment Required

- ADALM-PLUTO SDR
- GNU Radio Companion (GRC)
- Antenna
- USB Cable
- Computer/Laptop

Software Requirements

- GNU Radio
- PlutoSDR Drivers
- libiio Support

Pulse Shaping and Matched Filter System

The above flowgraph generates random QPSK symbols and applies pulse shaping using a Root Raised Cosine filter. The filtered signal is observed in both constellation and frequency domains before transmission through PlutoSDR.

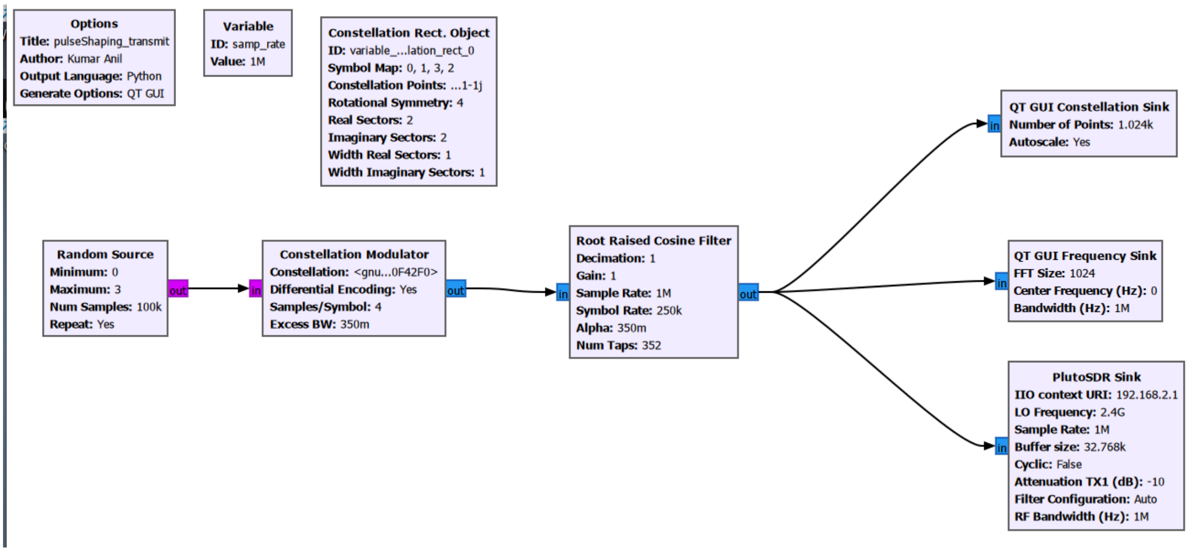

Figure 1: QPSK Pulse Shaping using Root Raised Cosine Filter

Flowgraph Description

Random Source

Generates random digital symbols for transmission.

QPSK Modulator

Maps the input symbols into QPSK constellation points.

Root Raised Cosine Filter

Performs pulse shaping and limits the occupied bandwidth of the transmitted signal. The filter also reduces inter-symbol interference.

QT GUI Constellation Sink

Displays the constellation points of the pulse-shaped signal.

QT GUI Frequency Sink

Displays the spectral characteristics of the filtered signal.

PlutoSDR Sink

Transmits the pulse-shaped QPSK signal over the air.

Working Principle

The Random Source generates digital symbols which are mapped into QPSK constellation points by the Constellation Modulator. These symbols are then passed through a Root Raised Cosine filter which shapes the transmitted pulses and limits spectral spreading.

The pulse-shaped signal is monitored using constellation and frequency sinks before being transmitted using PlutoSDR. At the receiver, an identical Root Raised Cosine filter can be used as a matched filter to maximize signal recovery and minimize inter-symbol interference.

Parameters

Parameter	Value
Sample Rate	1 MS/s
Modulation Scheme	QPSK
Samples per Symbol	4
Symbol Rate	250 kSymbols/s
Roll-off Factor (α)	0.35
Number of Taps	352
Carrier Frequency	2.4 GHz
RF Bandwidth	1 MHz

Table 1: Pulse Shaping Parameters

Procedure

1. Open GNU Radio Companion.
2. Create a Random Source block to generate digital symbols.
3. Add a QPSK Constellation Modulator.
4. Configure the Root Raised Cosine filter for pulse shaping.
5. Connect QT GUI Constellation Sink and Frequency Sink for visualization.
6. Configure PlutoSDR Sink for wireless transmission.
7. Run the flowgraph.
8. Observe the constellation diagram and transmitted spectrum.
9. Analyze the bandwidth reduction achieved by pulse shaping.

Observation

The Root Raised Cosine filter successfully reduced spectral spreading and limited the occupied bandwidth of the QPSK signal. The constellation remained stable while the transmitted spectrum became smoother and more bandwidth efficient.

Result

Pulse shaping using a Root Raised Cosine filter was successfully implemented. The filter reduced inter-symbol interference and improved spectral efficiency while maintaining proper QPSK symbol representation.

Applications

- Digital Communication Systems
- QPSK and QAM Modems
- Satellite Communication
- Software Defined Radio Systems
- Wireless Communication Networks

Conclusion

The experiment demonstrated pulse shaping using a Root Raised Cosine filter in a QPSK communication system. The filtered signal occupied less bandwidth and exhibited reduced inter-symbol interference, improving overall transmission quality.

References

1. GNU Radio Documentation
2. Analog Devices PlutoSDR Documentation
3. Digital Communication Systems by Simon Haykin
4. Software Defined Radio Principles and Applications